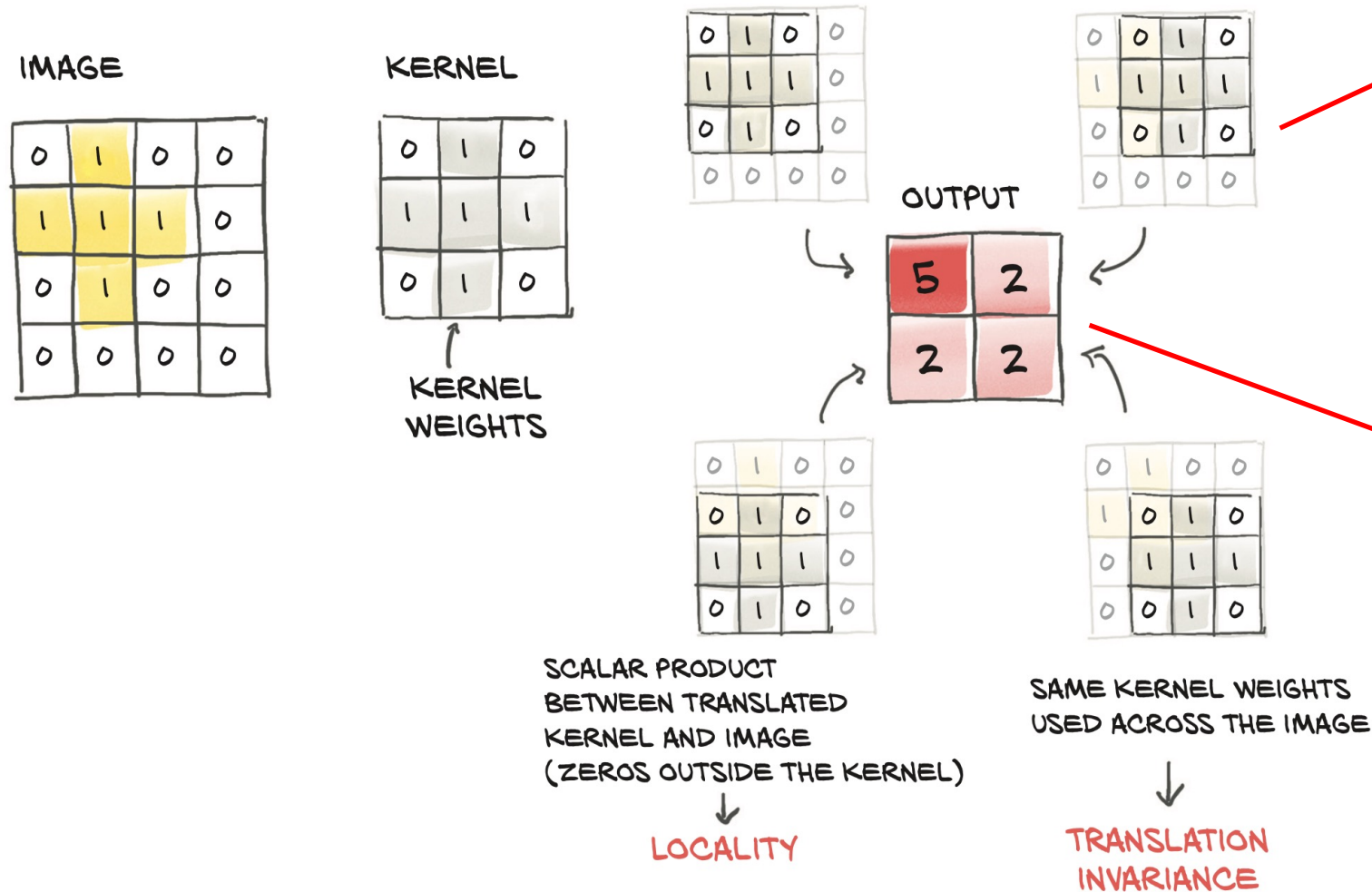


Lecture 10: Deep Learning For Sequential Data

Convolution: Recap

- The convolutional neural network (CNN) is a specialized type of neural network model designed for working with image data.
 - Convolutional layer performs a linear operation called a convolution: the multiplication between input data and an array of weights, called a filter or a kernel.

Convolution: Recap



- An element-wise multiplication between the image and filter/kernel, and then is summed, resulting in a single number/value.
- The filter is smaller than the input image, which allows the same filter/kernel to be applied to each filter-size patch, left to right, top to bottom.

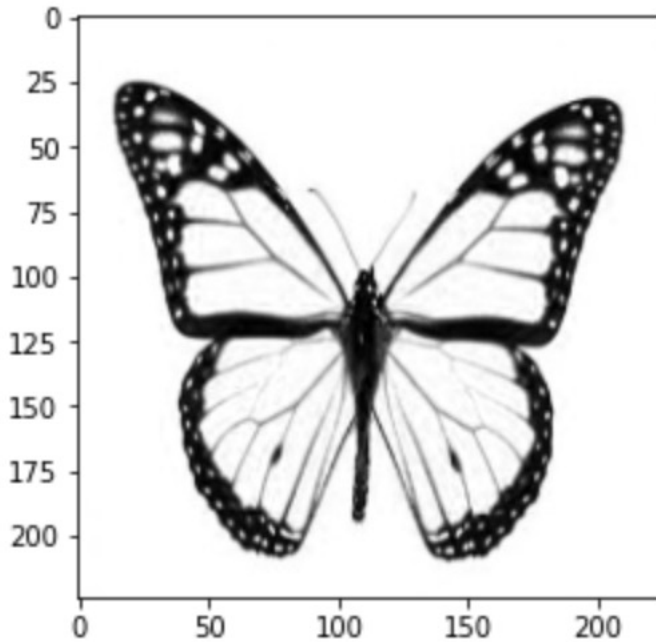
<https://machinelearningmastery.com/convolutional-layers-for-deep-learning-neural-networks/>

Convolution: Recap

- A convolution is equivalent to having multiple linear operations whose weights are zero almost everywhere except around individual pixels.
 - Models with a lot fewer parameters: the number of parameters depends not on the number of pixels in the image, but rather on the size of the convolution kernel and on how many convolution filters.
 - Local operations on neighborhoods
 - Translation invariance

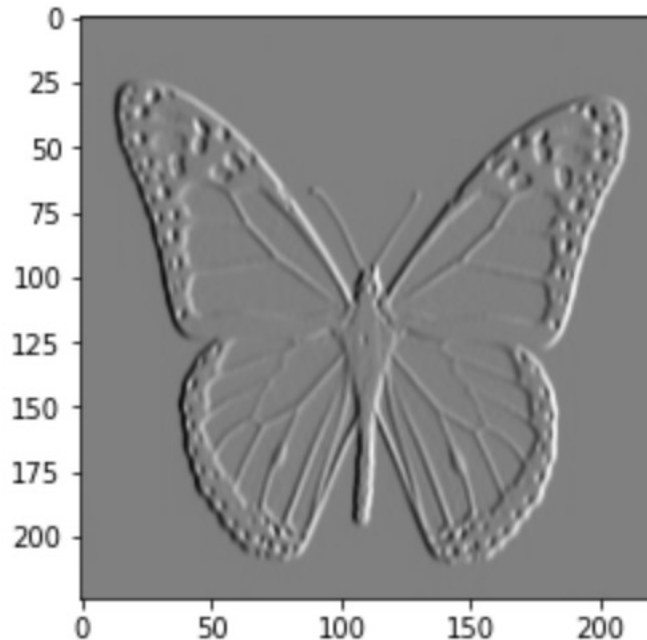
Detecting Features with Convolutions: Recap

■ Translation invariance



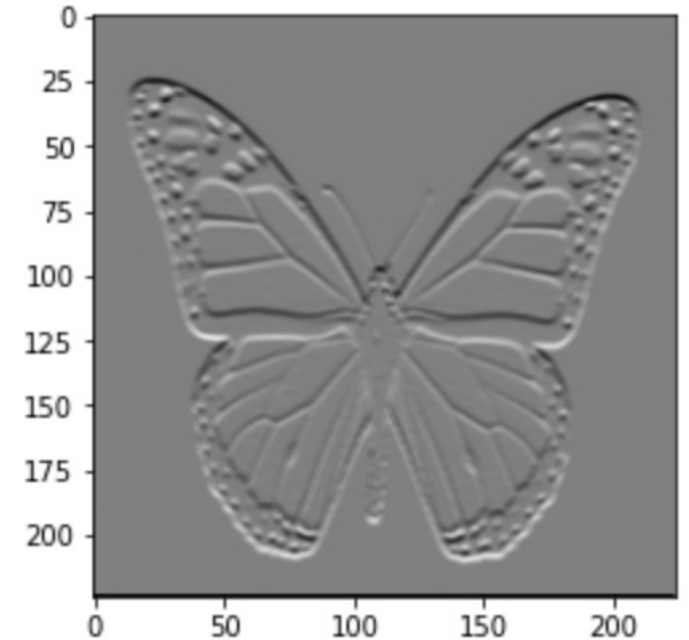
The convolution kernel enhances the vertical edges

-1	0	1
-1	0	1
-1	0	1



The convolution kernel enhances the horizontal edges

-1	-1	-1
0	0	0
1	1	1

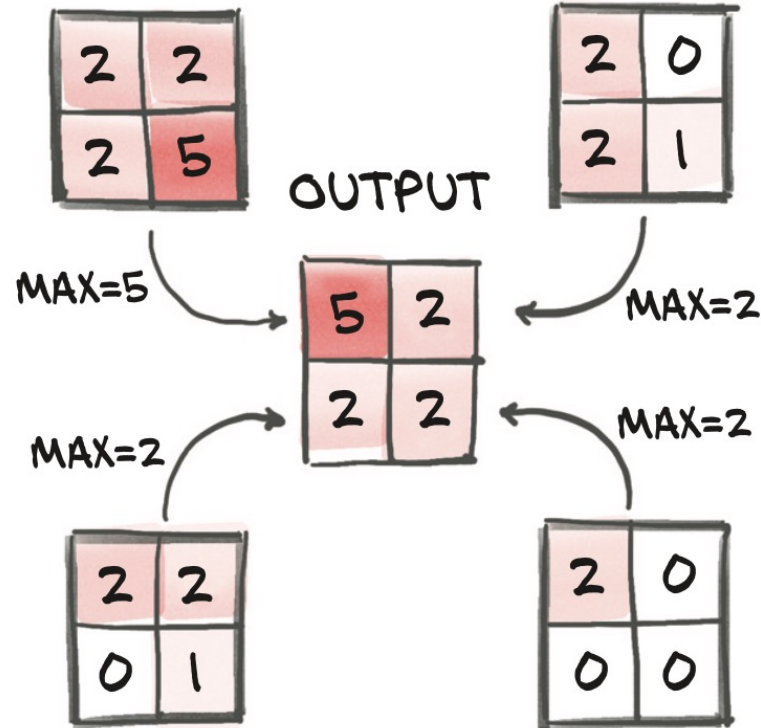


Looking Further with Depth and Pooling: Recap

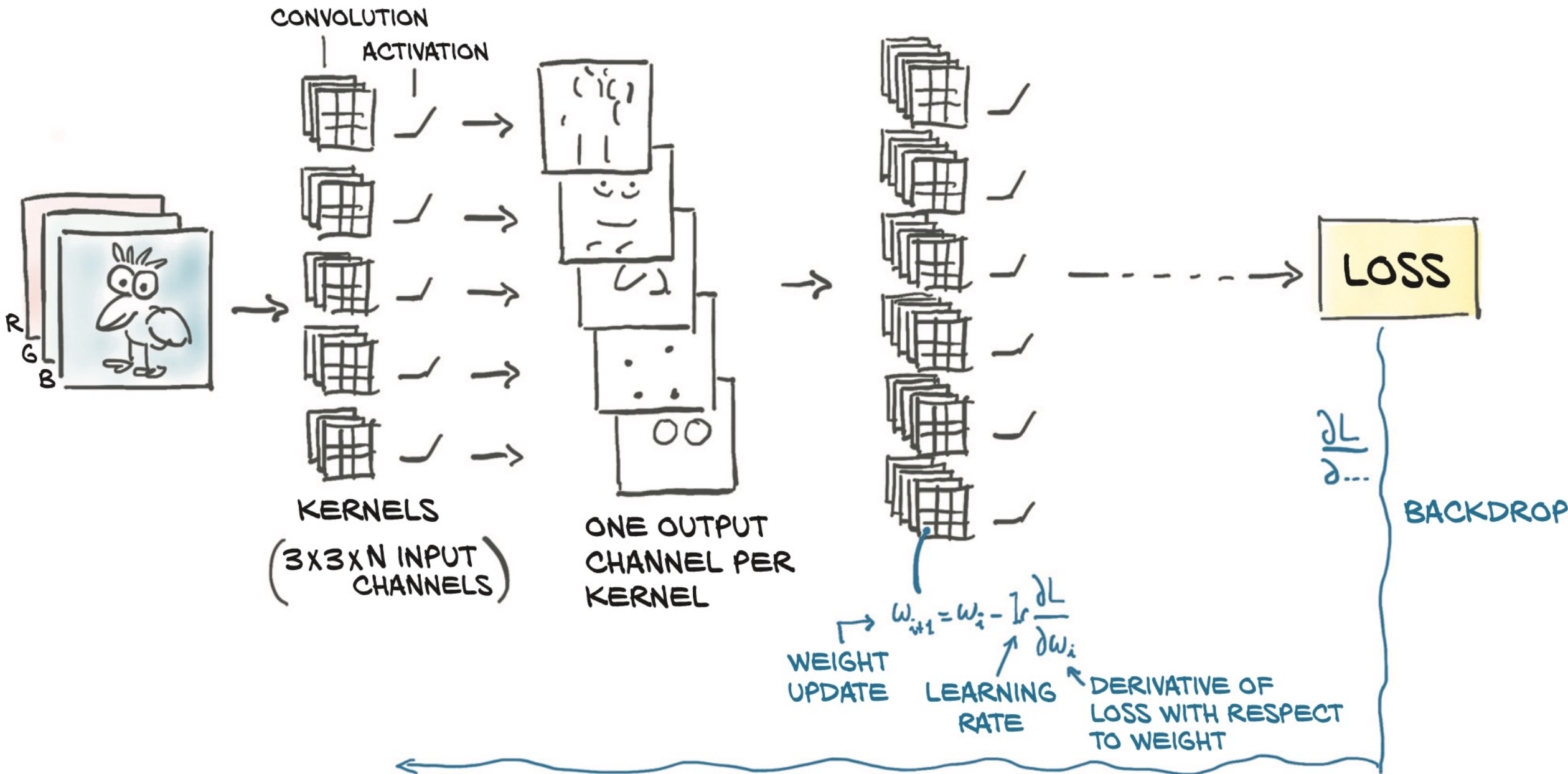
INPUT
(OUTPUT OF CONV + ACTIVATION)

2	2	2	0
2	5	2	1
2	2	2	0
0	1	0	0

MAXPOOL



- By keeping the highest value in the 2×2 neighborhood as the downsampled output, we ensure that the features that are found *survive* the downsampling.

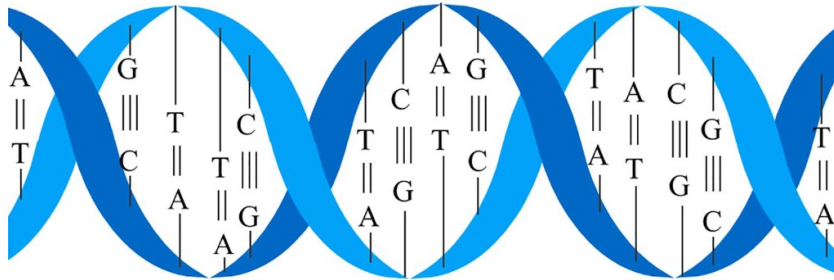


Overview

- Sequential Data
- RNN: Recurrent Neural Network
- Pizza restaurant sales

Sequential Data

- Sequential Data refers to **any data that contain elements that are ordered into sequences.**
 - Time series: stock market, company sales, patient data
 - Ordered: text, DNA sequences



DNA sequence

order_date ▼	order_time ▼	unit_price ▼	total_price ▼
1/1/15	11:38:36	13.25	13.25
1/1/15	11:57:40	16	16
1/1/15	11:57:40	18.5	18.5
1/1/15	11:57:40	20.75	20.75
1/1/15	11:57:40	16	16
1/1/15	11:57:40	20.75	20.75
1/1/15	12:12:28	16.5	16.5
1/1/15	12:12:28	20.75	20.75
1/1/15	12:16:31	16.5	16.5
1/1/15	12:21:30	16.5	16.5
1/1/15	12:29:36	12.75	12.75
1/1/15	12:29:36	12	12

Pizza restaurant sales

Translate What You See through Your Camera

- Google Translate mobile app has the instant camera translation, which allows you to see the world in your language by just pointing your camera lens at the foreign text.

How Google translate works?

- Google's AI translation methods is known as neural machine translation (NMT).
- These models have been incorporated into Google Lens and the web version of Translate, but they are now supporting instant camera translation.



Predict Stock Prices through Natural Language Processing

- Early in 1998, Wuthrich et al. attempted to predict stock markets based on online news articles such as The Wall Street Journal.
- Now, the natural language processing community correlates the text contents with the sentiment (positive or negative) of financial forums posts to predict the stock prices.



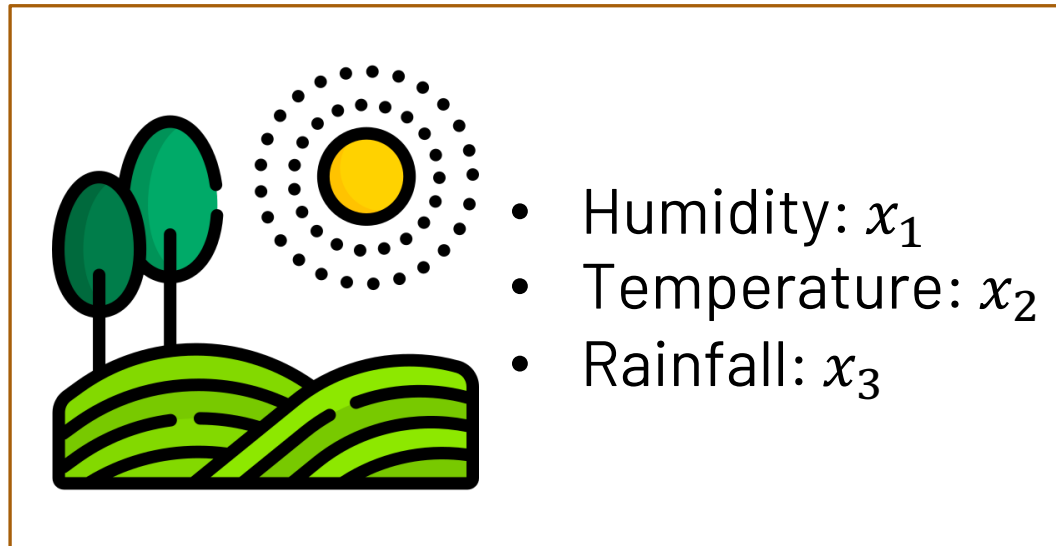
Another type of study is about applying social media text mining for stock market prediction.

- The method leverages market information contained in high-volume social media data rather than news articles.

Sequential Data

- Why is sequential data an issue for the fully connected neural network?
 - The model cannot remember the data
 - The data points are dependent on the other data points in the same sequence

Sequential Data

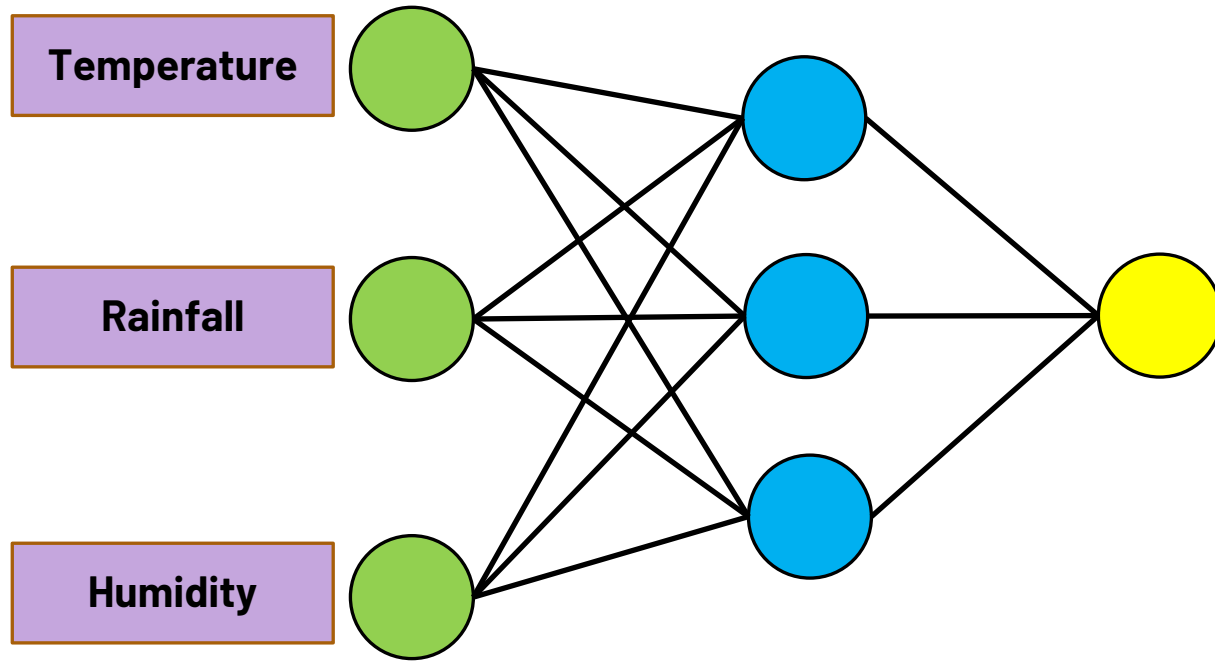


predict



The Yield of Citrus

Sequential Data



Date	temperature	rainfall	humidity
2022-06-01	85	12	89
2022-06-02	89	9	88
...			
2022-12-20	67	120	67

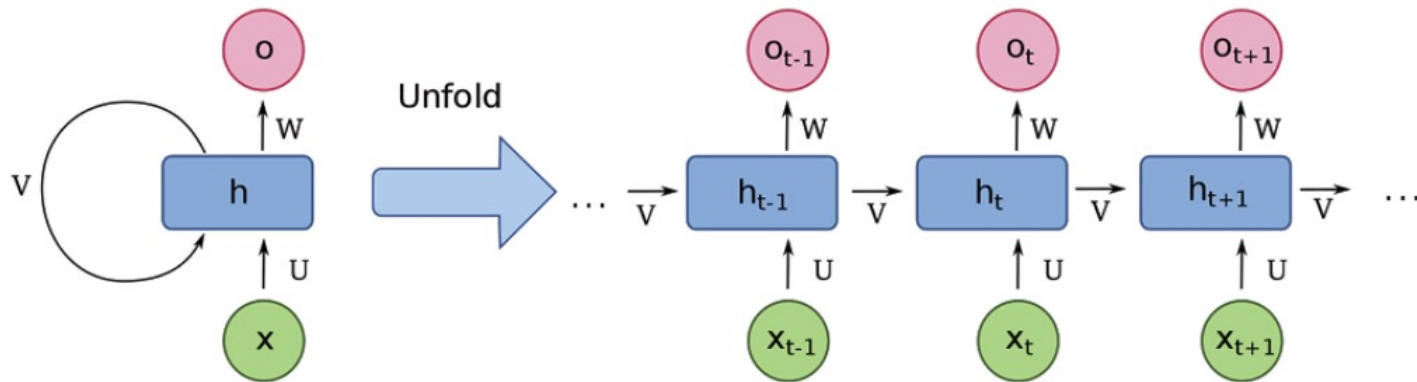
Yield of Citrus

- The subsequent yield of citrus depend on the long-term historical temperature, rainfall, and humidity.
- The traditional neural network assumes the data is non-sequential, and that **each data point is independent of other data points**. It does **not remember the previous data** that is important to predict the yield.

Recurrent Neural Network

Recurrent neural network (RNN): also known as RNNs, are a class of neural networks that allow previous outputs to be used as inputs while having hidden states.

Recurrent neural network

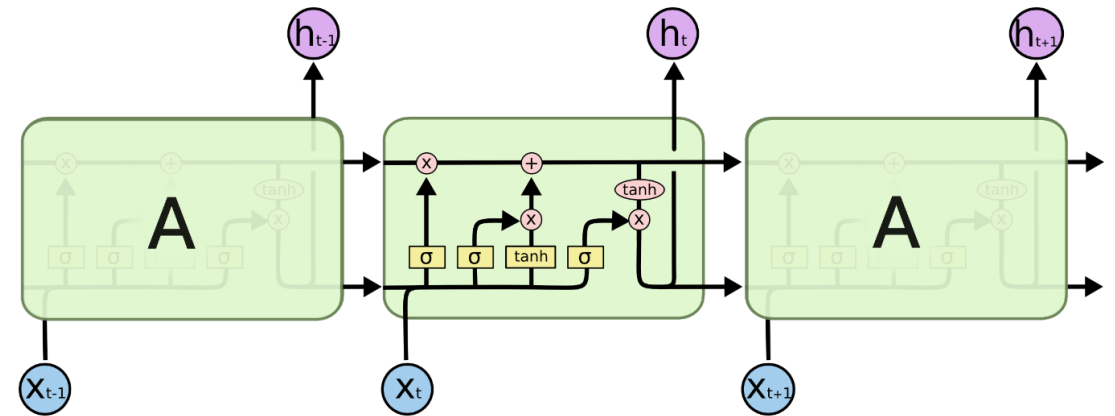


- The idea behind RNN is to make use of sequential information
- Before we assume all inputs are independent of each other
- RNN performs the same task for every element of sequence, with the output being depended on the previous computations.
- Have the “memory” which captures information about what has been calculated so far

Recurrent Neural Network

- The recurrent neural network has a mechanism that can handle the sequential dataset.

Long short-term memory (LSTM) is a special kind of RNN, capable of learning long-term dependencies.



Pizza Restaurant Sales



DATASET:

<https://www.kaggle.com/datasets/shilongzhuang/pizza-sales>

Let's predict the future total sales of pizza based on the sequential transactions by using LSTM.